

MDP Fundamentals

MDP Tuple: $(\mathcal{S}, \mathcal{A}, P, r, \lambda)$

- $\mathcal{S}$: finite state space
- $\mathcal{A}$: finite action space
- $P(s'|s, a)$: transition probabilities
- $r(s, a)$: reward function
- $\lambda \in [0, 1)$: discount factor

Policies: Markov $\pi_t : \mathcal{S} \rightarrow \mathcal{A}$; Stationary $\pi : \mathcal{S} \rightarrow \mathcal{A}$; History-dependent $\pi_t : H_t \rightarrow \mathcal{A}$
Theorem 1 (Adequacy of Markov Policies). *For infinite horizon: $\exists$ stationary Markov policy that is optimal*

Value: $V^\pi(s) = \mathbb{E}_\pi [\sum_{t=0}^\infty \lambda^t r(s_t, a_t) | s_0 = s]$
Q-Function: $Q^\pi(s, a) = r(s, a) + \lambda \sum_{s'} P(s'|s, a) V^\pi(s')$

Finite Horizon

Bellman Equation:

$$V_t(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s'} P(s'|s, a) V_{t+1}(s') \right]$$

Boundary: $V_T(s) = 0$ (or terminal reward)
Optimal Policy:

$$d_t^*(s) = \arg \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s'} P(s'|s, a) V_{t+1}(s') \right]$$

Infinite Horizon - Discounted

Bellman Operator: $T : V \mapsto TV$ where

$$(TV)(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s'} P(s'|s, a) V(s') \right]$$

Theorem 2 (Contraction). *T is a contraction with modulus λ :*

$$\|TV - TU\|_\infty \leq \lambda \|V - U\|_\infty$$

Theorem 3 (Banach Fixed Point). *Unique fixed point V^* exists: $V^* = TV^*$*
Optimality Equation:

$$V^*(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s'} P(s'|s, a) V^*(s') \right]$$

Theorem 4 (Optimal Policy Existence). *$\exists$ stationary deterministic policy π^* s.t.*

$$V^{\pi^*}(s) = V^*(s) \quad \forall s$$

Value Iteration

Algorithm:

1. Initialize $V_0(s) = 0 \quad \forall s$
2. For $n = 0, 1, 2, \dots$:

$$V_{n+1}(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s'} P(s'|s, a) V_n(s') \right]$$

3. Stop when $\|V_{n+1} - V_n\|_\infty < \varepsilon$
4. Extract policy: $d(s) = \arg \max_a [r(s, a) + \lambda \sum_{s'} P(s'|s, a) V_{n+1}(s')]$

Convergence: $\|V_n - V^*\|_\infty \leq \frac{\lambda^n}{1-\lambda} \|V_1 - V_0\|_\infty$

Stopping Criterion: $\|V_{n+1} - V_n\|_\infty < \frac{\varepsilon(1-\lambda)}{2\lambda} \Rightarrow \|V_{n+1} - V^*\|_\infty < \varepsilon$

Gauss-Seidel Value Iteration

Algorithm:

1. Initialize $V_0(s) = 0 \quad \forall s$
2. For $n = 0, 1, 2, \dots$:
3. For each $s \in \mathcal{S}$ (in order):

$$V_{n+1}(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \lambda \sum_{s' < s} P(s'|s, a) V_{n+1}(s') + \lambda \sum_{s' \geq s} P(s'|s, a) V_n(s') \right]$$

Advantage: Uses updated values immediately (faster convergence)

Policy Iteration

Algorithm:

1. Initialize policy d_0
2. **Policy Evaluation:** Solve $(I - \lambda P^{d_k})V = r^{d_k}$
3. **Policy Improvement:** $d_{k+1}(s) = \arg \max_a [r(s, a) + \lambda \sum_{s'} P(s'|s, a) V^{d_k}(s')]$
4. If $d_{k+1} = d_k$, stop; else go to step 2

Splitting Method: $(I - \lambda P^d) = (I - \lambda Q_d) - \lambda R_d$ Solve: $V = (I - \lambda Q_d)^{-1} (r^d + \lambda R_d V)$ iteratively

Theorem 5 (Policy Improvement). $V^{d_{k+1}} \geq V^{d_k} \quad \forall s$, strict if d_k not optimal

Average Reward MDP

Average Reward:

$$g^\pi = \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} r(s_t, a_t) \right]$$

Assumptions: Unichain + irreducible under all policies

Bias Function:

$$h^\pi(s) = \lim_{N \rightarrow \infty} \mathbb{E}_\pi \left[\sum_{t=0}^{N-1} (r(s_t, a_t) - g^\pi) | s_0 = s \right]$$

Theorem 6 (Unichain Optimality Equations). *$\exists$ scalar g^* and function h^* s.t.*

$$g^* + h^*(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \sum_{s'} P(s'|s, a) h^*(s') \right]$$

Optimal policy: $d^*(s) = \arg \max_a [r(s, a) + \sum_{s'} P(s'|s, a) h^*(s')]$

Bellman Error:

$$B(g, h)(s) = \max_{a \in \mathcal{A}} \left[r(s, a) - g + \sum_{s'} P(s'|s, a) h(s') - h(s) \right]$$

Policy optimal iff $B(g, h)(s) = 0 \quad \forall s$

Relative Value Iteration

Algorithm:

1. Fix reference state s_0 , initialize $V_0(s) = 0$
2. For $n = 0, 1, 2, \dots$:

$$V_{n+1}(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \sum_{s'} P(s'|s, a) V_n(s') \right] - V_n(s_0)$$

3. $g^* = \lim_{n \rightarrow \infty} V_n(s_0)$
4. $h^*(s) = \lim_{n \rightarrow \infty} (V_n(s) - V_n(s_0))$

Policy Evaluation

Discounted: $V^\pi = (I - \lambda P^\pi)^{-1} r^\pi$

Average Reward:

- Stationary dist: π^π solves $\pi^\pi P^\pi = \pi^\pi, \sum_s \pi^\pi(s) = 1$
- Gain: $g^\pi = \sum_s \pi^\pi(s) r(s, \pi(s))$
- Fundamental matrix: $Z = (I - P^\pi + P^*)^{-1}$ where $P^* = \mathbf{1} \pi^\pi$
- Bias: $h^\pi = Z(r^\pi - g^\pi \mathbf{1})$

Key Proof Ideas

Contraction Proof:

$$|(TV)(s) - (TU)(s)| \leq \lambda \max_{s'} |V(s') - U(s')| = \lambda \|V - U\|_\infty$$

Policy Improvement:

$$V^{d_{k+1}} \geq T V^{d_k} = T V^{d_k} \geq V^{d_k}$$

Optimality: $V^* = T V^* \Rightarrow V^* \geq V^\pi$ for all π

Important Formulas

Discounted Value:

$$V^\pi(s) = \sum_{t=0}^\infty \lambda^t \mathbb{E}_\pi [r(s_t, a_t) | s_0 = s]$$

Action Value:

$$Q^*(s, a) = r(s, a) + \lambda \sum_{s'} P(s'|s, a) V^*(s')$$

Policy Gradient (if needed):

$$\nabla_\theta V^\pi(s) = \sum_{s'} \sum_{t=0}^\infty \lambda^t P^\pi(s_t = s' | s) \sum_a \nabla_\theta \pi(a | s') Q^\pi(s', a)$$

Convergence Rate:

$$\|V_n - V^*\|_\infty \leq \frac{\lambda^n}{1-\lambda} \|V_1 - V_0\|_\infty$$

Structural Results

Theorem 7 (Monotonicity). *If $r_1(s, a) \leq r_2(s, a) \quad \forall (s, a)$, then $V_1^*(s) \leq V_2^*(s)$*

Theorem 8 (Convergence as $\lambda \rightarrow 1$). *For unichain MDP: $\lim_{\lambda \rightarrow 1^-} (1-\lambda) V_\lambda^*(s) = g^*$*

Modified Policy Iteration

Algorithm:

1. Initialize V_0 , policy d_0
2. Policy evaluation: $V_{k+1} = T^{d_k} V_k$ (one step)
3. Policy improvement: $d_{k+1} = \arg \max_a T^a V_{k+1}$
4. Repeat until convergence

Computational Complexity

- Value Iteration: $O(|S|^2|\mathcal{A}|)$ per iteration
- Policy Iteration: $O(|S|^3)$ per evaluation
- Gauss-Seidel: Same complexity, faster convergence
- Relative VI: $O(|S|^2|\mathcal{A}|)$ per iteration

Multichain MDPs

Classification:

- **Unichain:** Every policy induces single recurrent class

- **Multichain:** Some policies have multiple recurrent classes

Theorem 9 (Multichain Optimality Equations). $\exists$ gain vector g^* and bias h^* s.t.:

$$g^*(s) + h^*(s) = \max_{a \in \mathcal{A}} \left[r(s, a) + \sum_{s'} P(s'|s, a) h^*(s') \right]$$

$$g^*(s) = \max_{a \in \mathcal{A}_s^*} \left[\sum_{s'} P(s'|s, a) g^*(s') \right]$$

where $\mathcal{A}_s^* = \{a : g^*(s) + h^*(s) = r(s, a) + \sum_{s'} P(s'|s, a) h^*(s')\}$

Multichain Policy Eval: Solve $(I - P^\pi)h^\pi = r^\pi - g^\pi, (I - P^\pi)g^\pi = 0$

Error Bounds

Value Iteration:

$$\|V_n - V^*\|_\infty \leq \frac{\lambda^n}{1 - \lambda} \|V_1 - V_0\|_\infty$$

Policy Iteration:

$$V^{d_{k+1}} - V^{d_k} \geq (I - \lambda P^{d_k})^{-1} (T^{d_{k+1}} - T^{d_k}) V^{d_k}$$

Special Cases

Separable MDP: $r(s, a) = r_S(s) + r_A(a), P(s'|s, a) = P_A(s'|a) \Rightarrow d^*(s) = \arg \max_a [r_A(a) + \sum_{s'} P_A(s'|a) h^*(s')]$ (constant policy)

Linear Programming

Primal LP: $\min \sum_s V(s)$ s.t. $V(s) \geq r(s, a) + \lambda \sum_{s'} P(s'|s, a) V(s') \forall s, a$

Dual LP: $\max \sum_{s,a} r(s, a) x(s, a)$ s.t. $\sum_a x(s', a) = \delta_{s_0}(s') + \lambda \sum_{s,a} P(s'|s, a) x(s, a), x(s, a) \geq 0$

Optimal: V^* from primal = optimal value function

Convergence Theorems

Theorem 10 (Value Iteration Convergence). $\lim_{n \rightarrow \infty} V_n = V^*$ and $\lim_{n \rightarrow \infty} d_n = d^*$

Theorem 11 (Policy Iteration Convergence). *Policy iteration converges in finite steps to optimal policy*

Theorem 12 (Gauss-Seidel Convergence). *Gauss-Seidel VI converges to V^* (faster than standard VI)*

Q-Learning

$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \lambda \max_{a'} Q(s', a') - Q(s, a)]$ Convergence: $\sum_t \alpha_t = \infty, \sum_t \alpha_t^2 < \infty$

Norms

$$\|V\|_\infty = \max_s |V(s)|, \|V\|_w = \max_s \frac{|V(s)|}{w(s)}, \text{sp}(V) = \max_s V(s) - \min_s V(s)$$

Existence & Uniqueness

Theorem 13 (Optimal Policy Exists). *Finite MDP $\Rightarrow \exists$ deterministic stationary optimal policy*

Theorem 14 (Unique Value). *V^* is unique fixed point of T*

Theorem 15 (Unique Gain). *Unichain MDP $\Rightarrow g^*$ unique*

Key Inequalities

Bellman: $V^\pi(s) \leq r(s, \pi(s)) + \lambda \sum_{s'} P(s'|s, \pi(s)) V^\pi(s')$ Optimality: $V^*(s) = \max_a [r(s, a) + \lambda \sum_{s'} P(s'|s, a) V^*(s')]$

Tips

Sparse matrices; Vectorize: $V_{n+1} = \max_a [r + \lambda P V_n]$; Early stop; Normalize by ref state

Stationary Dist

$\pi^\pi P^\pi = \pi^\pi, \sum_s \pi^\pi(s) = 1$. Solve: $(P^\pi)^T - I$ with $\sum_s \pi^\pi(s) = 1$

Performance Bounds

Suboptimal: $\|V^\pi - V^*\|_\infty \leq \frac{1}{1-\lambda} \max_s |(T^\pi V^*)(s) - V^*(s)|$ Approx: If $\|V - V^*\|_\infty < \varepsilon$, then $\|V^\pi - V^*\|_\infty < \frac{2\lambda\varepsilon}{1-\lambda}$